



Can Large Language Models forecast carbon price movements? Evidence from Chinese carbon markets

Rui Chen^a, Haiqi Jiang^b, Tingyu Guo^a, Chenyou Fan^{b,*}

^a International Business College, South China Normal University, Guangzhou, Guangdong, China

^b School of Artificial Intelligence, South China Normal University, Guangzhou, Guangdong, China

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ABSTRACT

This paper investigates the impact of Large Language Models (LLMs) on forecasting Chinese carbon prices. We introduce a novel two-stage forecasting framework integrating a Time-Series Model (TSM) and Large Language Models. Initially, we use historical data on Chinese Emission Allowance prices to train the TSM for preliminary predictions. LLMs then refine these predictions, which process a sequence of past and corresponding future prices as a chain of thought. Additionally, we utilize the LLM to analyze and categorize the sentiment of news headlines, generating market sentiment labels that enhance the LLM's predictive accuracy. Our findings indicate that LLMs can improve TSM forecasts by 28–38 % across different regional markets. Furthermore, incorporating news sentiment labels into the LLM contributes an additional reduction in forecasting deviations, ranging from 3–4 %.

1. Introduction

The recent proposal of carbon neutrality aims to eliminate net carbon emissions in the next 20 to 30 years. To regulate economic activities towards this goal, countries like China and the European Union have established carbon markets where emission allowances can be traded. Industrial manufacturers can either purchase more allowances or reduce their emissions, promoting the use of cleaner energy and encouraging innovation.

Accurately predicting carbon prices can help manufacturing companies minimize costs through effective planning, while also offering valuable insights to governments for regulating domestic industrial sectors. The challenge in predicting the carbon market arises from the lack of sufficient data in novel markets, as well as the presence of non-linearity and volatility, rendering traditional prediction methods less effective. To tackle this challenge, we propose enhancing carbon price forecasting by integrating machine-learning-based Time-Series Models with recent advancements in Pre-trained Large Language Models in the field of Artificial Intelligence.

In recent years, the progress of Artificial Intelligence (AI) such as Machine Learning has sparked emerging research into their impact on financial research, including market trend prediction (Gu et al., 2020), energy price forecasting (Chai et al., 2023), financial textual analysis (Loughran and McDonald, 2016), and corporate governance (Gofman and Jin, 2024). Especially, the recent emerging Large Language Models (LLMs) have been proven to possess robust knowledge-learning capability (Brown et al., 2020; Ouyang et al., 2022). Through pre-training on extensive human knowledge, LLMs acquire a rich understanding not only of human languages but also

* Corresponding author.

E-mail address: fanchenyoun@scnu.edu.cn (C. Fan).

of mathematical deduction and reasoning (Wei et al., 2022), as well as financial marketing (Sun et al., 2021; Zhou and Wang, 2021; Li and Liu, 2023). In our study, we employ LLMs to improve the accuracy of the Time-Series Model estimates and to analyze financial sentiment in news headlines related to carbon emissions. We aim to address the following critical research questions: Can LLMs comprehend the price trends within the Chinese carbon emission market? Do LLMs possess the capability to enhance the prediction outcomes of conventional machine learning models? Can LLMs effectively leverage sentiment signals from news headlines to forecast prices? To answer these questions, we develop a two-stage time-series forecasting framework integrating both a Time-Series Model and a Large Language Model.

In our paper, we gather data on Chinese Emission Allowance (CEA) prices from 2015 to 2022 in China's three regional carbon markets. We collect macro-economic influencing factors as predictors such as commodity prices of oil and coal, as well as stock indices regarding cleaner energy (Ren et al., 2022). We then pre-process the data with missing data filling and feature selection using a traditional Lasso-based method (Pang et al., 2023). Next, we use a deep-learning-based Time-Series Model to fit temporal patterns of historical CEA prices in a standard supervised way using historical data. For this purpose, we employ the state-of-the-art Autoformer as our TSM implementation (Wu et al., 2021) and utilize the well-trained Autoformer model to generate raw future predictions based on the learned market context patterns condensed in the model parameters.

To explore our research questions, we leverage the capabilities of the Large Language Models to incorporate market context, world knowledge, and associations of regional markets into predictions. We first attempt to directly prompt the LLMs to predict the future price trend of the carbon market by providing past historical steps. Unfortunately, we observed that the prediction results of the direct prompting approach underperform the Autoformer model, which means that the LLMs fail to understand the future price trend by directly prompting.

Thus, we propose to use LLMs to predict carbon price trends based on the past prediction results of the Autoformer models, that is, employing LLMs to refine the Autoformer's prediction results. This refinement is achieved by providing the LLM with exemplary pairs of TSM's raw predictions and actual future price sequences, prompting it to learn from past prediction deviations. This approach is supported by the following reasons. Firstly, Autoformer predictions have already incorporated market context for LLMs reference and are expected to help LLMs to better understand the carbon market trend. Secondly, Autoformer predictions suffer from past deviations as all Time-Series Models, while LLMs possess the ability to continuously learn from new data through fine-tuning, enabling them to adapt to evolving past price trends and enhance their prediction accuracy over time. Such advanced LLM prompting techniques are called Chain-of-Thought (CoT) and Self-Refine in the Artificial Intelligence area. The CoT technique enables successful predictions based on observable historical patterns, such as preceding periods, thereby providing the LLM with relevant market context. The Self-Refine procedure further encourages the LLM to actively reflect on past TSM predictive inaccuracies and refine its current predictions accordingly. Inspired by this, we propose to first utilize CoT to incorporate market context into future predictions provided by the LLM, followed by further prompting the LLM to refine its predictions based on reflection from past deviations. Therefore, our two-stage framework, incorporating the Time-Series Model and Large Language Models, enables the LLMs to account for market-context variations and yield forecast results more accurately than Autoformer models and direct prompt processes. Furthermore, after adding noise to the TSM prediction results inputted into the LLM, we find that only more accurate TSM results enable the LLM to make more accurate predictions. This indicates that the LLM is indeed capable of identifying carbon market price trends and further learning and refining from the TSM predictions.

Next, we assess the LLM's capability for textual sentiment analysis by creating a sentiment label dataset of news headlines related to Chinese climate policy and carbon emissions events sourced from major news media. For each headline, we use LLM to assess whether it is positive, negative, or neutral for future carbon prices. We convert these responses into numerical scores and integrate them into LLM prediction. Additionally, we investigate whether incorporating such market sentiment data can enhance LLM's ability to forecast carbon market price trends.

Overall, we conducted a comprehensive series of empirical analyses to show that our CoT-RF methodology reduce the prediction MSE by 28–38 %, by reflecting on deviations of TSM predictions over preceding time steps. Moreover, by utilizing news sentiment labels as additional information provided to the LLM, we observe another 3–4 % improvement in our LLM-based methods. Our results show that increased model complexity improves not only information processing but also market context and sentiment comprehension. This suggests that utilizing an advanced TSM-LLMs time-series framework has the potential to significantly improve forecast accuracy. The observed pattern of CEA price predictability based on recent public news sentiment supports theories of delayed information diffusion and limited investor attention (Kahneman, 1973).

Our study contributes to the literature in several ways. First, our paper is among the first to investigate the application of LLMs for predicting price movements in China's carbon emissions allowances market. While recent research has employed LLMs to forecast stock price movements (Lopez-Lira and Tang, 2023) and to interpret managerial expectations regarding corporate policies (Jha et al., 2024), our study extends this approach to the emerging field of carbon markets. We have compiled a comprehensive dataset of CEA prices across different regions, along with a range of economic factors influencing these prices. By demonstrating the potential of LLMs in enhancing the ability of market participants to process information and predict trends in carbon markets, our study highlights their value for both market forecasting and decision-making. Achieving more accurate carbon price forecasts can support more rational carbon quota allocations, foster the sustainable development of global carbon markets, and facilitate progress toward the goal of carbon neutrality.

Second, our paper is related to the recent strand of the literature on the Fintech area that leverages machine learning techniques to address finance-related research questions (Hoberg and Phillips, 2016; Gu et al., 2020; Cao et al., 2023; Jiang et al., 2023). We propose a novel two-stage framework that combines a state-of-the-art Time-Series model for initial predictions with LLMs to enhance and refine the forecasts. A key feature of our approach is the implementation of a self-reflection mechanism, enabling the LLM to learn from

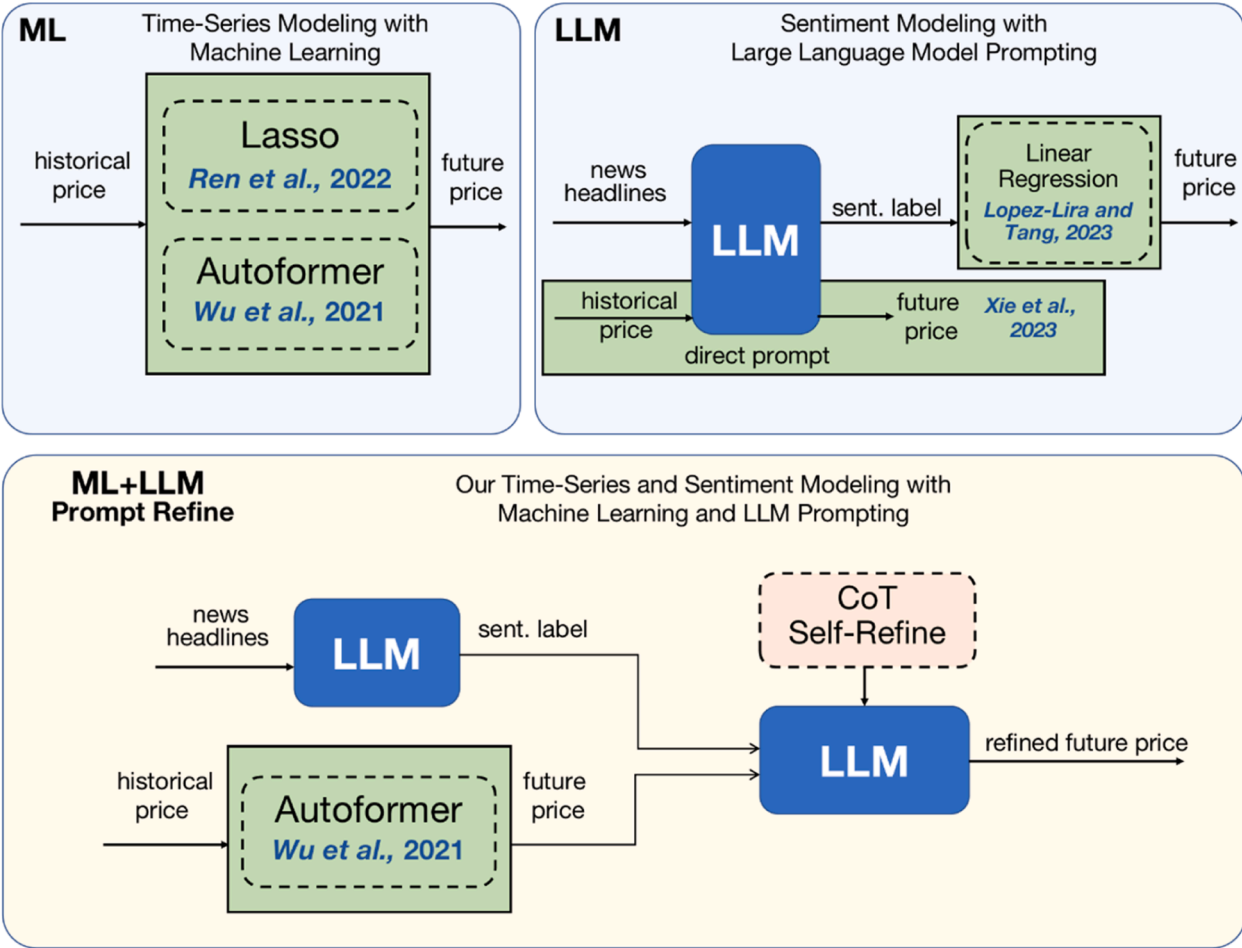


Fig. 1. Motivation for incorporating both Machine Learning (ML) and Large Language Model (LLM) in carbon market forecasting.

previous forecasting errors, thereby improving future predictions.

Finally, our study contributes to the literature that employs linguistic analysis of news articles to extract sentiment for market prediction (Loughran and McDonald, 2016). Financial sentiment analysis is widely used in financial market price forecasting. We differ from prior studies by utilizing LLMs for sentiment analysis of public news, allowing the sentiment extracted to further refine the accuracy of carbon price predictions generated by machine learning models. We demonstrate that LLMs can effectively analyze complex information, such as future trends in carbon emission markets, and show that this information is both interpretable and actionable for market participants.

2. Literature review

2.1. Carbon market prediction

As the issue of carbon emissions has gained increasing attention, research has focused on enhancing the accuracy of carbon price predictions using advanced artificial intelligence algorithms. Previous studies primarily use time-series models, such as ARIMA or GARCH, for price forecasting, demonstrating that advanced machine learning techniques can significantly enhance prediction accuracy. Zhu (2012) introduces a general framework for carbon price prediction based on decomposition and synthesis methods. Zhou et al. (2022) advanced this by decomposing multi-dimensional data to capture both long-term trends and short-term fluctuations. More recently, Ren et al. (2022) use Quantile Group Lasso for feature selection and carbon futures price prediction in the European Union Emissions Trading System. Pang et al. (2023) propose the use of adaptive sparse Quantile Group Lasso to achieve more robust price predictions. Duan et al. (2023) use an attention mechanism-based Long Short-Term Memory neural network to forecast carbon futures price dynamics under the EU Emission Trading Scheme. While existing studies have explored various machine learning methods, there is a gap in integrating these techniques with more recent advancements in artificial intelligence. For instance, leveraging large language models or hybrid models combining different AI approaches could enhance predictive accuracy. Furthermore, research on price prediction for China's carbon emission market is limited and often focuses solely on the early markets of Beijing and Guangdong. Our study addresses this gap by analyzing data from carbon markets in three regions of China and further examining the effectiveness of our methodology in learning novel markets based on the knowledge of existing markets.

2.2. Financial sentiment analysis

Financial sentiment analysis has been deeply applied in the financial field due to the breakthrough of machine learning algorithms. Previous studies utilize textual analysis to analyze the level of financial sentiment based on word frequency (Loughran and McDonald, 2014). Xing et al. (2020) study the common mistakes of applying classical shallow deep-learning models for financial sentiment analysis. FinBERT (Araci, 2019; Yang et al., 2020) is proposed to classify sentiment labels of finance related texts. Xie et al. (2023) find that ChatGPT is not capable of directly prediction stock price movements. Lopez-Lira and Tang (2023) use GPT to extract sentiment labels of news headlines, categorizing them as positive, negative, or neutral, and apply these labels in a linear regression model to forecast stock prices, demonstrating a significant improvement in prediction accuracy. Glasserman and Lin (2023) highlight the critical role of incorporating financial news sentiment into stock price prediction, finding that it has a more substantial impact than the inherent biases related to company names. Therefore, investigating the impact of market sentiment on carbon price predictions could provide additional market information and a more comprehensive understanding of price dynamics.

2.3. Large language models (LLMs)

Recently, LLMs such as the GPT family and LLaMA show great advantages in modeling various language tasks, including arithmetic reasoning and question answering (Brown et al., 2020; Ouyang et al., 2022; Touvron et al., 2023). Several studies propose to simulate the human thinking process by LLMs, such as thinking step-by-step with Chain-of-Thought (Wei et al., 2022), reflecting on experience (Shinn et al., 2023) and making further refinement over past decisions such as Self-refine (Madaan et al., 2023). We will show the effectiveness of improving CEA forecasts by exploring LLM's semantic understanding and self-refine capabilities. Some studies also explore the use of LLMs for financial tasks, such as finance-related content generation and question answering (Li et al., 2023; Wu et al., 2023), extracting information from corporate policies (Jha et al., 2024), and mining trading signals or factors (Chen et al., 2023; Wang et al., 2023).

Overall, Fig. 1 illustrates the motivation for incorporating both Machine Learning (ML) and LLM for carbon market forecasting. The top-left box presents studies using ML methods, such as Lasso and Autoformer, for time-series prediction. The top-right box shows recent works of prompting LLM with sentiment labels or historical prices for future predictions. The bottom box outlines our methodology, which involves using Autoformer for initial prediction and then refining these outputs by prompting the LLM. We also apply recent advanced prompting techniques, such as Chain-of-Thought and Self-Refine, to further enhance forecast performance.

3. Data and classical machine learning methodology

We first introduce data and features used in the Chinese Carbon Emission Allowance (CEA) forecasting task. Then we demonstrate our multi-stage learning framework.

Table 1
Summary statistics for carbon price across different market zones.

Summary	Hubei	Shenzhen	Guangdong
Period	05/01/2015–11/06/2022	04/05/2015–11/06/2022	04/05/2015–11/06/2022
N	2595	2476	2476
Min	10.38	1.00	8.10
Max	61.48	49.99	95.26
Mean	25.13	23.95	22.47
Median	25.55	25.17	17.77
Std.dev	8.04	11.14	11.43
Q1	16.95	13.65	14.33
Q3	29.41	32.95	27.98

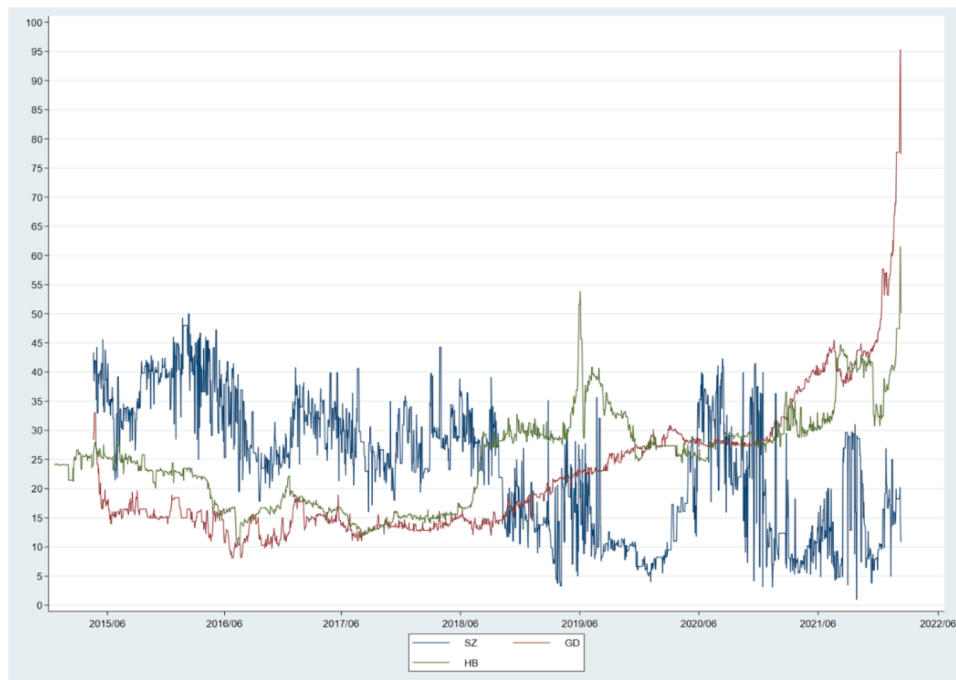


Fig. 2. Carbon market pricing trends over the sample period. This figure plots the time series price trend of Shenzhen (SZ, blue line), Guangdong (GD, red line) and Hubei (HB, green line) carbon markets from June 2015 to June 2022.

3.1. Data and feature selection with lasso-based method

We collect the CEA closing price (P_{close}), trade volume (Vol), and trade amount (Amt) each day for three regional markets, including Hubei (HB), Shenzhen (SZ) and Guangdong (GD), from the Wind database. Our sample period is from 2015 to 2022. The summary statistics of the carbon price in each market are shown in Table 1. During the sample period, we observed that the standard deviation of carbon prices in the Guangdong market was notably the highest among the three regions. In February 2022, the trading price of carbon emissions in the Guangdong market reached a peak of 95 yuan, surpassing the prices observed in the HB and SZ markets. Fig. 2 illustrates the temporal trends in carbon market pricing across the sample period. The Hubei market illustrated by the green line shows consistent growth across the entire timeline but experiences two distinct peaks at approximately late 2017 and early 2021. The Shenzhen market is represented by the blue line, we observe that its prices exhibit moderate fluctuations throughout most of the observation period. However, a notable increase is observed around mid-2020, followed by another sharp rise toward the end of the study period. These trends likely reflect the enforcement of stricter carbon reduction policies and growing attention to carbon trading during these times. In contrast, the Guangdong market depicted by the red line shows more consistent growth across the entire timeline. Its prices steadily climb without any major dips, indicating stable demand and continuous participation in the market. Overall, while each region exhibits unique characteristics in their respective carbon market pricing dynamics, all show upward trajectories reflecting the general trend of increasing interest and investment in carbon reduction initiatives.

In addition, we collect auxiliary economic data, including the China ETF volatility index (VXFXI), China Daqing crude oil spot price (P_{oil}), and China Qinhuangdao coal spot price (P_{coal}) from the Wind and CNRDS database. Detailed descriptions are presented in Table 2, Panel A. Following previous work (Ren et al., 2022), we obtain 23 carbon economy stock indices from China's stock market,

Table 2
List of variables.

Variable	Definition
Panel A: Numerical features with descriptions.	
P_{close}	The closing price of CEA.
Vol	The trading volume of CEAs, measured by the total number of units exchanged.
Amt	The total trade amount of CEAs, representing the monetary value of all transactions.
VXFXI	The China ETF Volatility Index as global market sentiment.
P_{oil}	The spot price of Daqing crude oil in China.
P_{coal}	The spot price of Qinhuangdao coal in China.
Panel B: Selected Carbon Indices of various CEA Zones	
Hubei	Carbon Tech 60, Mainland L-C Index
Shenzhen	Carbon Tech 60, Mainland L-C Index
Guangdong	Carbon Tech 30 & 60, Mainland L-C Index

This table reports the variables utilized for feature selection. Panel A shows the variable description of selected features from each CEA market. Panel B shows the selected market carbon indices for different CEA zones.

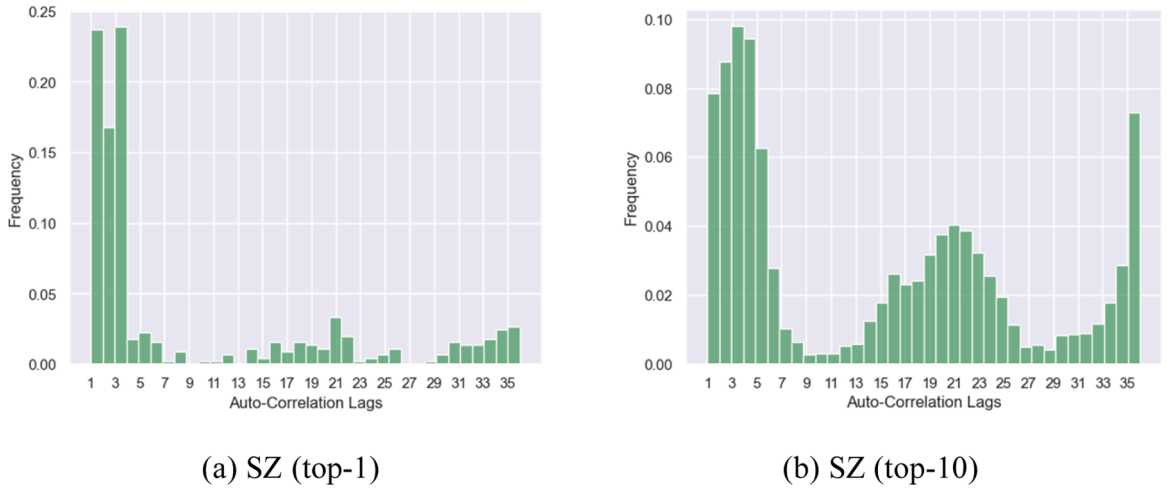


Fig. 3. Statistics of learned lags for the Shenzhen market using the Autoformer method. This figure plots the top-1 and top-10 lags from all the test cases of the Shenzhen (SZ) carbon market. The X-axis represents the lag in terms of a day, while the Y-axis indicates the frequency of time series with a specific lag.

each having its values of closing price, volume, and total traded amount. Subsequently, we employ the classical Lasso method for feature selection to identify the most relevant stock indices for predicting carbon price fluctuations within a specific CEA zone. We regress the future carbon price on all stock indices and extract the coefficients from the Lasso regression model. Indices with non-zero coefficients are retained for each CEA zone, and from these, we further select the top 2 or 3 most relevant features, provided their coefficient scores exceed a threshold of 0.1. Table 2, Panel B presents the details of the selected carbon indices features for different CEA zones. The *Mainland L-C Index* is the CSI Mainland Low-Carbon Economy index, which is composed of 50 stocks in China's A-share market involving clean energy power generation, energy conversion and storage, cleaner production and consumption, and waste disposal. The *Carbon Tech 60* represents the CNI CIKD Carbon Neutral Technology Power Index, which includes 60 stocks from the Shenzhen Stock Exchange in the A-share market as sample stocks. The *Carbon Tech 30* is the SZSE ChiNext Carbon Neutral Technology Power Index, consisting of 30 stocks selected from the Growth Enterprise Market of the Shenzhen Stock Exchange. The index compilation is based on the classification of listed companies within the carbon-neutral technology service industry.

3.2. Time-series modeling with autoformer (AF)

For each zone and each period, we observe the past T_h days as $T_h = \{1, 2, \dots, T_h\}$ and predict the subsequent T_f future days as $T_f = \{T_f + 1, \dots, T_h + T_f\}$ for the CEA forecasting task. In this paper, we utilize Autoformer (Wu et al., 2021) as the state-of-the-art backbone Time-Series Model based on the Transformer deep-learning Architecture to provide initial coarse carbon price forecasts.

The Autoformer has two main components. The first one is Auto-Correlation, which is a mechanism that can capture the period-based dependencies of historical price (X_h) by computing the correlation of subseries with different time delays and aggregating to a new series denoted by χ . The second component is Series-Decomposition, which takes the auto-correlated series χ and separates them into trend-cyclical $\mathcal{T}$ and seasonal parts $\mathcal{S}$. In the Autoformer model, the Auto-Correlation and Series Decomposition processes are iteratively applied. The final future prediction ($\hat{Y}$) is generated by combining the last trend and seasonal components derived from

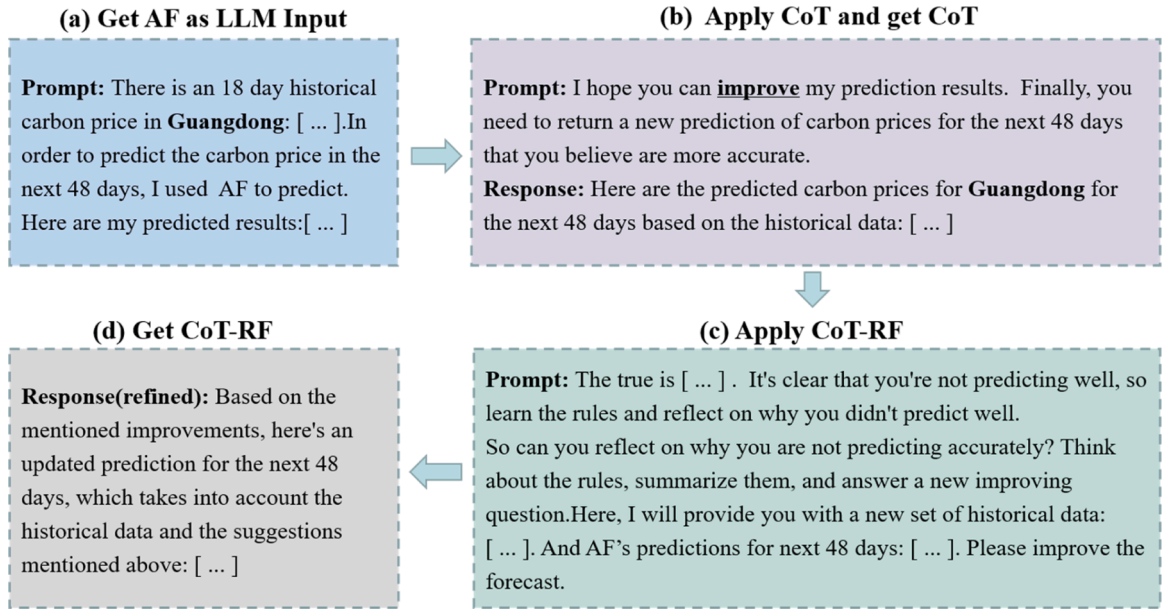


Fig. 4. Illustration of the CoT refinement pipeline with specific prompts and responses. This figure illustrates the application of the Chain-of-Thought (CoT) refinement pipeline with specific prompts and responses. Step (a): The trained Autoformer (AF) model predicts future prices, resulting in AF. Step (b): The LLM CoT is applied to enhance the AF predictions, yielding CoT results. Step (c): The LLM is further prompted to self-refine the future predictions. Step (d): The refined results are obtained, denoted as CoT-RF.

these repeated iterations, such as, $\hat{Y} \leftarrow \mathcal{S} + \mathcal{T}$. We train the Autoformer to fit the model parameters on the CEA data. The trained model serves as our TSM, providing future predictions, denoted as $\hat{Y}^{AF}$, based on an arbitrary sequence of historical price, X_h , as illustrated in Eq. (1):

$$\hat{Y}^{AF} \leftarrow AF(X_h). \quad (1)$$

The auto-correlation of time series χ_t with a lag τ indicates the time-delay similarity between χ_t and its τ lagged series $\chi_{t-\tau}$. In our case, the unit of the time lag is one day. We iterate $\tau \in [1, 2, \dots, L]$ and calculate the top-1 and top-10 most correlated lags for the carbon price data. In Fig. 3, we visualize the histogram of the top-1 and top-10 most correlated lags of the Shenzhen market. The X-axis represents the lag with a unit of a day, while the Y-axis indicates the frequency of time series with a specific lag. Fig. 3(a) collects only the top-1 most correlated lag of each time series, while Fig. 3(b) collects the top-10 most correlated lags. We observe that the top-1 lags in Shenzhen are concentrated within the first few days, while the top-10 lags exhibit either short-term or long-term (three weeks) trends in Fig. 3(b). This pattern suggests the presence of volatility and non-linearity in the CEA markets.

4. Apply large language model for forecast refining

In this study, we utilize ChatGPT as the underlying large language model for carbon price forecasting of CEA markets based on in-context learning capacities.

4.1. Direct prompting LLM methodology

First, we introduce a straightforward method that directly prompts the LLM to predict the carbon price for future steps without using TSM. To this end, we instruct the LLM with prompts such as “Give the historical carbon price: [...], please predict the price for the next 48 days.” We extract the predicted sequence from the LLM’s response. We denote this method as Direct Prompting (DP).

The DP is also known as “zero-shot learning” in the field of artificial intelligence, as it relies entirely on the built-in knowledge of the LLM to predict the future (Brown et al., 2020). We adopt this method as a weak baseline in our empirical analysis. Our findings in Section 5 are consistent with previous work by Xie et al. (2023), indicating that current LLMs are insufficiently equipped to accurately predict market prices without being provided with adequate market context.

4.2. Chain-of-Thought methodology

In contrast to the DP, we introduce the Chain-of-Thought (CoT) methodology to enhance the capability of the LLMs to refine the prediction results. Our CoT method utilizes the TSM to interpret the historical data and produce raw future predictions first. Then we

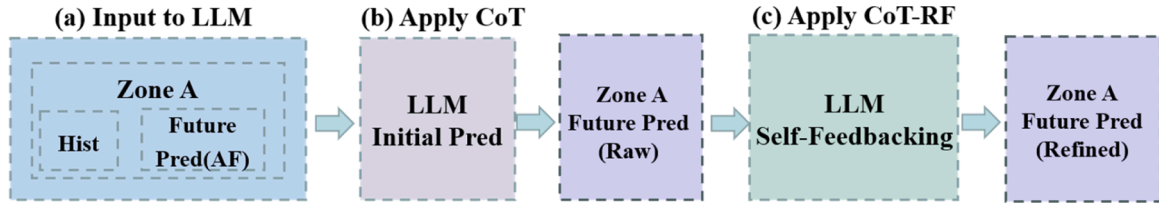


Fig. 5. Flowchart of refining Autoformer predictions using LLM with CoT and CoT-RF. This figure presents a flowchart outlining the process of refining Autoformer predictions using LLM with the Chain-of-Thought (CoT) and CoT refinement (CoT-RF) methods. Step (a): The trained Autoformer (AF) model is applied to predict future prices. Step (b): The LLM is used to enhance the AF predictions, resulting in the results of CoT. Step (c): The LLM is further prompted to self-refine the future predictions, producing results of CoT-RF.

prompt the LLM to directly refine the AF predictions. The corresponding textual prompt is designed as “We give you historical carbon price [...] and AF’s predicted price for the next 48 days [...]. Please improve AF’s prediction”. Fig. 4(a) and (b) display a typical CoT prompt process. We denote the predicted future prices after the CoT refining process as $(AF + CoT)_{\mathcal{T}_f}$.

4.3. Chain-of-Thought refinement methodology

We further improve the CoT by additionally introducing a feedback process to refine its answer, thus called CoT Refinement (CoT-RF). Compared to the CoT, we further provide the LLM with actual market prices and prompt it to reflect on its past errors. This process leverages the LLM’s contextual learning capabilities, enabling it to extrapolate from a limited set of examples (past Autoformer predictions and true outcomes) and generalize to new scenarios (future and refined Autoformer predictions).

The CoT-RF prompt is designed as “We give you some historical carbon prices [...] and AF’s predicted prices. The true sequence is [...]. Can you reflect on why you are not predicting accurately? Then I will provide you AF’s predictions for the next 48 days [...]. Please improve the forecast”. In Fig. 5, we illustrate the sequential process where Autoformer is used to generate initial predictions, followed by applying a chain-of-thought (CoT) method to enhance these predictions, and then executing reflection on the LLM’s response to further refine them (CoT-RF). This process is further elaborated with specific prompts in Fig. 4(a)–(d). We denote this sequential process as $(AF + CoT + RF)_{\mathcal{T}_f}$.

4.4. Sentiment analysis prompt

In this section, we collect a comprehensive news dataset for CEA markets through web scraping from five news media, including People’s Daily, Guangming Daily, China Economic Net, Securities Times, and 21st Century Business Herald. These are authoritative news media with wide readers in China ensuring their information is influential to participants in the carbon price market. People’s Daily and Guangming Online are the Chinese national central news media, while China Economic Net, Securities Times and 21st Century Business Herald are prominent financial media in China. The main contents we used for analysis are news headline and their release time. We begin by identifying keywords related to carbon pricing, such as “carbon emission permit” and “carbon emission”, from the news headlines and articles. Subsequently, we employ fuzzy search algorithms to query these keywords in the search engines of various websites, obtaining the relevant data through web scraping methods. Overall, after matching the news data, our final sample comprises 4391 headlines for financial sentiment analysis during the study period.

To guide the LLM in providing reasonable and logical responses about news sentiment, prompts are crucial. Following the study by Lopez-Lira and Tang (2023), we utilize ChatGPT for sentiment analysis and apply the following prompt to the publicly available headlines. The prompt consists of an instruction and input. In the instruction, we ask ChatGPT to pretend to be a financial expert with experience in carbon emissions trading. Given the necessity to accurately reflect the market sentiment conveyed by the news headlines, ChatGPT should be portrayed as a knowledgeable professional experienced in trading carbon emissions, likely possessing a keen understanding of carbon emission trading policies. According to the instructions, ChatGPT should give its analysis and its reason. For example, if the news headline has a potentially positive impact on carbon emission trading prices in the Chinese market, ChatGPT should answer “YES” in the first line and give a reason in the second line. The following example is based on a positive news headline.

Forget all your previous instructions. Pretend you are a financial expert about trading carbon emissions in Chinese Emission Allowance (CEA) markets. Answer “YES” if good news, “NO” if bad news, or “UNKNOWN” if uncertain in the first line. Then elaborate with one short and concise sentence on the next line. Is this headline good or bad for the short-term trend of carbon emission trading prices in the Chinese market?

[NEWS]: The first implementation cycle of the national carbon market was successfully concluded.

[Response]: YES

That’s the good news. The successful conclusion of the first compliance cycle of the national carbon market indicates that the market mechanism is working well and may enhance investors’ confidence in carbon futures trading, which is expected to push up the price of carbon emissions trading in the carbon emissions trading market.

Table 3
Performance of CEA Price Forecasting Using Different Methods.

	AF	DP	CoT	CoT-RF	CoT-Sent	CoT-Sent-RF
Panel A. HB						
MSE	1.815***	1.987***	1.343***	1.133***	1.283***	1.063***
Δ MSE	-	-0.172 (-9 %)	0.472 (26 %)	0.681 (38 %)	0.532 (29 %)	0.752 (41 %)
<i>t</i> -statistics		-0.3	1.69*	2.78*	1.72***	2.91***
Wilcoxon		11	80*	85***	81***	86***
Panel B. SZ						
MSE	29.066***	34.077***	23.889***	20.877***	21.409***	20.040***
Δ MSE	-	-5.011 (-17 %)	5.177 (18 %)	8.191 (28 %)	7.657 (26 %)	9.022 (31 %)
<i>t</i> -statistics		-1.91**	1.71***	1.67*	3.06*	3.00***
Wilcoxon		-63**	44***	40**	79***	77***
Panel C. GD						
MSE	7.754***	7.642***	5.488***	4.795***	5.012***	4.502***
Δ MSE	-	0.112 (1 %)	2.266 (28 %)	2.959 (38 %)	2.742 (35 %)	3.253 (42 %)
<i>t</i> -statistics		0.05	1.89*	1.98**	4.22***	3.66***
Wilcoxon		22	57*	62**	103***	97***

This table presents the **Mean Squared Error (MSE)** for forecasting performance across different models in markets, including Hubei (HB), Guangdong (GD) and Shenzhen (SZ). Δ MSE indicates the difference in MSE between the Autoformer (AF) and each alternative method, with percentage differences relative to AF shown in parentheses. The Autoformer (AF) model generates future predictions through supervised learning and serves as the baseline for comparison. The DP method uses the LLM to predict future prices directly based on historical data. CoT employs the LLM to learn patterns from complete prediction examples and refine AF predictions based on new historical data. CoT-RF, an extension of CoT, further prompts the LLM to adjust predictions by accounting for deviations from the actual future values of a given sequence. CoT-Sent and CoT-Sent-RF incorporate additional sentiment labels to guide the LLM in enhancing prediction accuracy. Statistical significance of Δ MSE is reported using *t*-statistics and Wilcoxon's signed-rank test, with ***, **, and * indicating significance at the 1 %, 5 %, and 10 % levels, respectively.

Input is the news headlines that we have collected. One news headline for each prompt so that ChatGPT can match the analysis with its reason. For each answer, we transform them into a numerical score called “Sentiment Score”, so that we can quantify the influence of news headlines. In our sample, the sentiment label on average is 0.198, with 1.409 positive labels as “+1”, 2239 UNKNOWN labels as “0”, and 743 negative labels “-1”. Thus, news headlines have an overall positive sentiment in our sample period, aligning with the upward trajectory of prices across all carbon emissions trading markets. In addition, we average the scores if there are multiple headlines for a specific zone on a given day. To ensure the stability and accuracy of sentiment classification from LLMs, we query the same news multiple times either within the same model or across different models. We then apply a majority voting rule to aggregate the responses, thereby enhancing the reliability of the sentiment analysis results.

We integrate sentiment labels into the CoT method as additional information and introduce this method as CoT-Sentiment (CoT-Sent). The corresponding textual prompt is designed as “We give you historical carbon price [...] and AF’s predicted price for the next 48 days [...]. In addition, the historical market sentiments are [...], with + 1 as positive and -1 as negative. Please improve AF’s prediction”. We expect that the LLM will leverage the latest public news sentiment to enhance the accuracy of future carbon emissions price forecasts based on CoT.

5. Empirical results

This section presents and discusses our empirical results. We introduce a range of methods for predicting the future CEA price in three different regions, Hubei (HB), Shenzhen (SZ), and Guangdong (GD). The first method, Autoformer (AF), serves as our baseline approach. AF involves training the Autoformer model using supervised learning techniques to generate future predictions. The second method, Direct Prompting (DP), entails utilizing the LLM to prompt the future price directly based on historical data. The third method, CoT, leverages the LLM to learn patterns from comprehensive prediction examples and subsequently enhances AF predictions using new observed historical data. CoT Refinement (CoT-RF) is our fourth method, which is based on further prompts the LLM to refine predictions by reflecting its deviations from the true future of a given sequence. Moreover, after considering market sentiment from carbon news, we integrate sentiment labels into CoT and introduce the fifth method CoT-Sentiment (CoT-Sent). CoT-Sent incorporates sentiment labels to prompt the LLM, enhancing predictions within the CoT framework. Lastly, the CoT-Sent-RF method builds upon CoT-Sent, leveraging sentiment labels to prompt the LLM and refine forecasting accuracy further.

5.1. Evaluation tasks and metrics

We evaluate all methods with the future regression task and the trend classification task by Mean Squared Error (MSE) and forecast accuracy (*Accuracy*). The MSE is calculated by averaging the squared errors over all future 30 predicted steps in the regression task. The lower the MSE, the better the prediction performance. Accuracy is the 3-way future trend classification task that evaluates the predicted price on Day-10, 20, and 30 as up, neutral or down, indicating the relative position at a future step compared with the mean

Table 4
CEA forecasting accuracy for Hubei (HB), Shenzhen (SZ), and Guangdong (GD) Regions.

Panel A. HB			
Method	Accuracy		
	Day-10	Day-20	Day-30
AF	25 %	40 %	20 %
DP	30 %	45 %	25 %
CoT	35 %	40 %	20 %
CoT-RF	50 %	45 %	30 %
CoT-Sent	55 %	50 %	50 %
CoT-Sent-RF	60 %	65 %	50 %
Panel B. SZ			
Method	Accuracy		
	Day-10	Day-20	Day-30
AF	35 %	40 %	35 %
DP	40 %	30 %	35 %
CoT	50 %	45 %	65 %
CoT-RF	50 %	60 %	75 %
CoT-Sent	55 %	45 %	60 %
CoT-Sent-RF	55 %	55 %	75 %
Panel C. GD			
Method	Accuracy		
	Day-10	Day-20	Day-30
AF	45 %	40 %	35 %
DP	35 %	50 %	30 %
CoT	60 %	75 %	60 %
CoT-RF	60 %	80 %	80 %
CoT-Sent	50 %	60 %	70 %
CoT-Sent-RF	55 %	75 %	65 %

This table presents the **Accuracy** of 3-way trend classification on Day-10, Day-20, and Day-30. The Autoformer (AF) model generates future predictions through supervised learning and serves as the baseline for comparison. The DP method uses the LLM to predict future prices directly based on historical data. CoT employs the LLM to learn patterns from complete prediction examples and refine AF predictions based on new historical data. CoT-RF, an extension of CoT, further prompts the LLM to adjust predictions by accounting for deviations from the actual future values of a given sequence. CoT-Sent and CoT-Sent-RF incorporate additional sentiment labels to guide the LLM in enhancing prediction accuracy. The best values for each column are presented in bold.

observed price p over the historical 18 steps. We define the neutrality class as a price range within $[(1-\alpha)p, (1+\alpha)p]$. The range of upward and downward classes are $((1+\alpha)p, +\infty)$ and $(-\infty, (1-\alpha)p)$, respectively. The α is a parameter that determines the width of the neutrality class around the reference price p and defines the thresholds for the upward and downward classes. We choose α to be 2 % in our analysis. For example, if the price on Day-10 is 36.0 and the mean historical price over the past window of 18 steps is 35.0, the increase is approximately 2.8 %, indicating an upward trend.

5.2. Forecasting result analysis

Table 3 presents the forecast results for each model across three market zones. The mean squared error (MSE) of each method is statistically significant across all markets. Among these, the Hubei market demonstrates the lowest MSE, while the Shenzhen market exhibits the highest. To compare the methods, we use the AF model as a baseline and calculate the difference in MSE between AF and other methods.

Our results show that the DP method fails to outperform the traditional deep learning-based AF model in both the Hubei and Guangdong markets and significantly underperforms AF in the Shenzhen market. Interestingly, while the LLM does not directly yield a substantially lower MSE in predicting carbon price trends, its application through the CoT and CoT-RF methods significantly improves the forecast accuracy of AF. Specifically, CoT-RF achieves an MSE reduction of 38 % (1.13 vs. 1.82) in the Hubei market, 38 % (4.80 vs. 7.75) in the Guangdong market, and 28 % (20.88 vs. 29.06) in the Shenzhen market. This demonstrates that CoT-RF consistently achieves state-of-the-art performance, reducing prediction MSE by 28–38 % by accounting for deviations in TSM predictions across preceding time steps.

Additionally, when evaluating the CoT-Sent and CoT-Sent-RF models, we observe a marked improvement in prediction performance compared to the traditional AF method. CoT-Sent achieves a significantly lower MSE, with reductions of 0.532 in the Guangdong market, 7.657 in the Shenzhen market, and 2.742 in the Hubei market. This suggests that LLMs add value to forecasting models by extracting pertinent information from news headlines and predicting carbon allowance market reactions. When sentiment labels from the news are integrated into the LLM and refinement process, CoT-Sent-RF produces the smallest MSE across all markets,

Table 5
Evaluation of LLM forecast refinement capabilities under noisy conditions.

Noise level	Original CoT-RF	5 %	10 %	20 %	30 %
Panel A. HB					
MSE	1.133***	1.177***	1.202***	1.348***	1.444***
ΔMSE	-	0.044 (3.89 %)	0.069 (6.08 %)	0.215 (18.96 %)	0.311 (27.27 %)
<i>t</i> -statistics		2.47**	2.54**	2.97***	3.08**
Wilcoxon		105***	105***	105***	105***
Panel B. SZ					
MSE	20.877***	22.785***	25.584***	26.641***	27.830***
ΔMSE	-	1.908 (9.13 %)	4.707 (22.54 %)	5.764 (27.61 %)	6.953 (33.30 %)
<i>t</i> -statistics		3.90***	4.74***	4.91***	4.77***
Wilcoxon		115***	115***	115***	115***
Panel C. GD					
MSE	4.795***	4.913***	5.633***	6.285***	6.811***
ΔMSE	-	0.118 (2.45 %)	0.838 (17.47 %)	1.490 (31.05 %)	2.016 (42.04 %)
<i>t</i> -statistics		3.09***	3.69***	3.67***	4.31***
Wilcoxon		105***	105***	105***	105***

This table presents the MSE values for forecasting performance across different noise levels in the Autoformer (AF) predictions using the CoT-RF methods. The ΔMSE indicates the difference in MSE for each noise level relative to the MSE of the original CoT-RF predictions, with percentage differences indicated in parentheses. The AF model trains the Autoformer to make future predictions through supervised learning. The CoT-RF method, an extension of CoT, further prompts the LLM to refine predictions by accounting for deviations from the actual future values of a given sequence. We report *t*-statistics and Wilcoxon's signed-rank test for ΔMSE, with ***, **, and * signifying significance at the 1 %, 5 %, and 10 % levels, respectively.

contributing to an additional 3–4 % reduction in forecasting error. These findings indicate that the LLM is capable of capturing and forecasting price fluctuations within China's carbon emission market with notable precision.

We further investigate whether LLM-based methods yield superior forecast accuracy in predicting market trends compared to the Autoformer model. Table 4 presents the results of predicting the price trend (rise, neutral, or fall) for Day-10, Day-20, and Day-30. Firstly, the AF and DP methods show no significant differences in performance. However, CoT-RF methods consistently outperform AF, with accuracy improvements ranging from 5 % to 45 %. Secondly, the incorporation of sentiment labels yields a substantial enhancement in forecast accuracy, particularly in the Hubei market as seen in Table 5, Panel A. Specifically, CoT-Sent achieves a 20 % improvement over CoT on Day-10 (55 % vs. 35 %), a 10 % improvement on Day-20 (50 % vs. 40 %), and a 30 % improvement on Day-30 (50 % vs. 20 %). Furthermore, CoT-Sent-RF surpasses CoT-RF by 10 % on Day-10 and by 20 % on both Day-20 and Day-30. Overall, the use of news sentiment data offers long-term predictive benefits in the Hubei market. However, this effect is not as pronounced in the Shenzhen and Guangdong markets, where the inclusion of sentiment labels does not consistently lead to improved accuracy.

5.3. Robustness tests

5.3.1. Evaluation of LLM forecast refinement capabilities under noisy conditions

The application of Large Language Models encounters challenges in addressing illusory uncertainties. To validate our methodologies and conclusions, we conduct a robust test to demonstrate that the LLM can effectively learn time-series patterns from the AF predictions and robustly refine them. In the CoT-RF methodology, we introduce random noise levels of 5 %, 10 %, 20 %, and 30 % into the AF predictions over similar forecasting periods. The LLM is then tasked with refining these predictions based on the noise inputs. If the LLM can accurately interpret price fluctuations in the carbon emissions market and improve upon AF's time-series predictions, then the performance of CoT-RF will degrade as noise increases, leading to greater deviations from true carbon prices. On the contrary, the LLM's outputs will be random, yielding no significant improvements in accuracy.

Using the original CoT-RF results as a benchmark, we report the corresponding increases in MSE for the noisy AF prediction variants in Table 5. The results show that the MSE values for the noisy AF predictions in the CoT-RF method are significantly higher than those of the original CoT-RF method across all markets. This indicates that the more accurate the AF Evaluation of LLM Transfer-Learning Capabilities Across Carbon Markets predictions, the better the CoT-RF results refined from those predictions.

In sum, our robustness tests confirm that the LLM is capable of learning from and refining the predictions made by the Time-Series Model, thereby enhancing the accuracy of AF's predictions.

5.3.2. Evaluation of LLM transfer-learning capabilities across carbon markets

In this section, we present an innovative transfer-learning approach that leverages the market context of a source zone to enhance the prediction of a target zone, without requiring access to the target zone's historical training process. Our approach assumes that the source zone possesses a mature CEA market with a comprehensive history, while the target zone represents a newly established exchange. The transfer-learning process is structured as follows: (1) The LLM receives input that includes historical data and future predictions by AF from source zone A, and historical data from target zone B. (2) The LLM then predicts the future data for Zone B

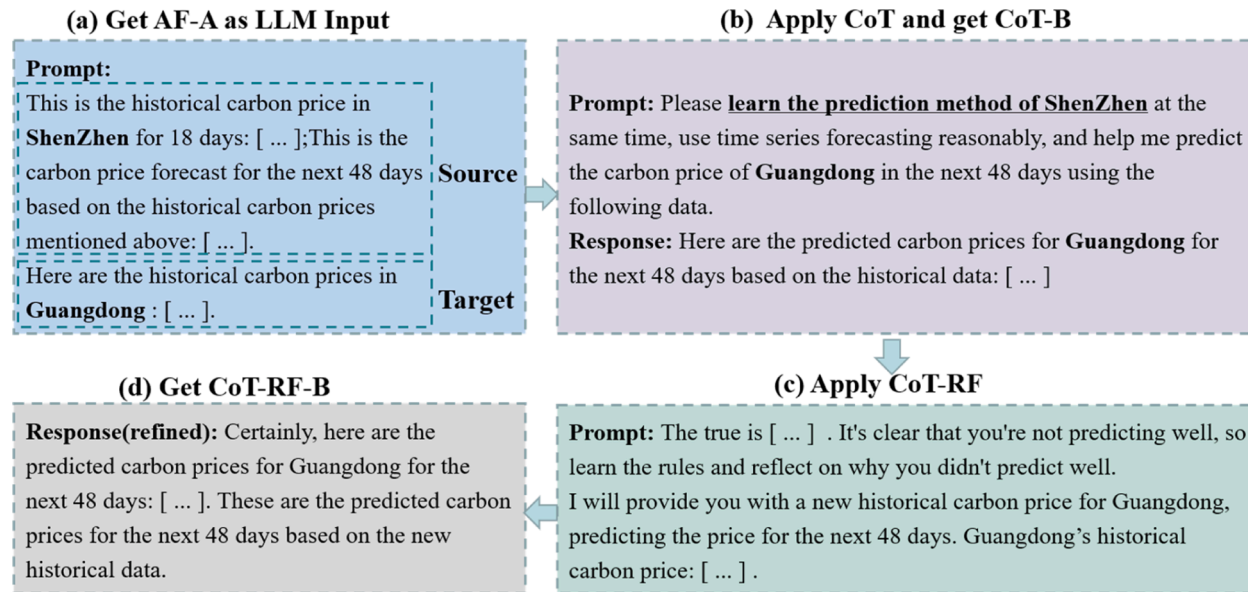


Fig. 6. Illustration of the Transfer-Learning based on CoT refinement pipeline with specific prompts and responses. This figure illustrates the Transfer-Learning process from the source market to the target market. Step (a): The LLM receives input from Autoformer that includes historical data and future predictions (AF-A) from source zone A, and historical data from target zone B. Step (b): The LLM then predicts the future data for Zone B based on its learning from the predictions in the source template and obtain CoT-B. Step (c): We prompt the LLM to self-refine future predictions. Step (d): The refined results are obtained, denoted as CoT-RF-B.

Table 6

Evaluation of LLM transfer-learning capabilities across different markets.

	AF	DP	CoT	CoT-RF
MSE	23.432***	28.828***	17.744***	13.188***
Δ MSE	-	-5.395 (-23 %)	5.688 (24 %)	10.244 (43 %)
t-statistics		0.39	2.46**	7.822***
Wilcoxon		47	67**	93***

This table presents the Mean Squared Error (MSE) for forecasting performance across Transfer-Learning based prediction in various models. Δ MSE indicates the difference in MSE between the Autoformer (AF) and each alternative method, with percentage differences relative to AF shown in parentheses. The AF model generates future predictions through supervised learning and serves as the baseline for comparison. The DP method uses the LLM to predict future prices directly based on historical data. CoT employs the LLM to learn patterns from complete prediction examples and refine AF predictions based on new historical data. CoT-RF, an extension of CoT, further prompts the LLM to adjust predictions by accounting for deviations from the actual future values of a given sequence. CoT and CoT-RF are based on the source zone's AF prediction. We report t-statistics and Wilcoxon's signed-rank test for Δ MSE, with ***, **, and * signifying significance at the 1 %, 5 %, and 10 % levels, respectively.

based on its learning from the predictions in the source template and obtain CoT-B. (3) We prompt the LLM to self-refine future predictions as CoT-RF-B. We elaborate this process step-by-step as shown in Fig. 6. In our empirical analysis, the SZ market is utilized as the source zone and the GD market as the target zone, as SZ is one of the earliest CEA markets in China. Table 6 reports the results of the analysis. We find that the MSE of the CoT-RF method is the smallest, with a significant reduction of 10.244 compared to the transfer-learning prediction through AF, representing a 43 % improvement in MSE for the GD market. These findings suggest that the LLM is capable of performing one-shot transfer forecasting effectively, benefiting from its ability to interpret the global market context embedded in the source zone's trends.

Overall, our transfer-learning test not only validates the ability of LLMs to predict trends in carbon market prices but also highlights the benefit of leveraging LLMs for forecasting in novel markets with limited historical data by drawing on knowledge and information from established markets.

6. Conclusion

We introduce a framework that integrates supervised TSM with LLMs prompting to enhance the accuracy of future forecasts for CEA markets. By presenting TSM predictions alongside relevant market contexts to the LLM through textual prompts, we effectively improve forecasting outcomes by leveraging the LLM's inherent few-shot learning capabilities. Additionally, we validate the LLM's remarkable capacity to learn from past errors and refine its predictive abilities through self-reflection. Lastly, by incorporating additional sentiment labels extracted from news headlines into the prompts, we observe a significant enhancement in forecast accuracy. Our results remain robust across various levels of noisy validation tests. Lastly, our methodology underscores the potential of utilizing advanced LLMs for various financial tasks such as news comprehension, risk assessment, and cross-regional trend predictions.

CRediT authorship contribution statement

Chen Rui: Conceptualization, Data curation, Supervision, Writing – original draft, Writing – review & editing. **Guo Tingyu:** Data curation, Investigation, Methodology, Validation. **Jiang Haiqi:** Data curation, Methodology, Software, Writing – original draft. **Fan Chenyou:** Conceptualization, Formal analysis, Resources, Supervision, Validation, Writing – original draft.

Declaration of Generative AI and AI-assisted technologies in the writing process

This paper does not use AI in scientific writing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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